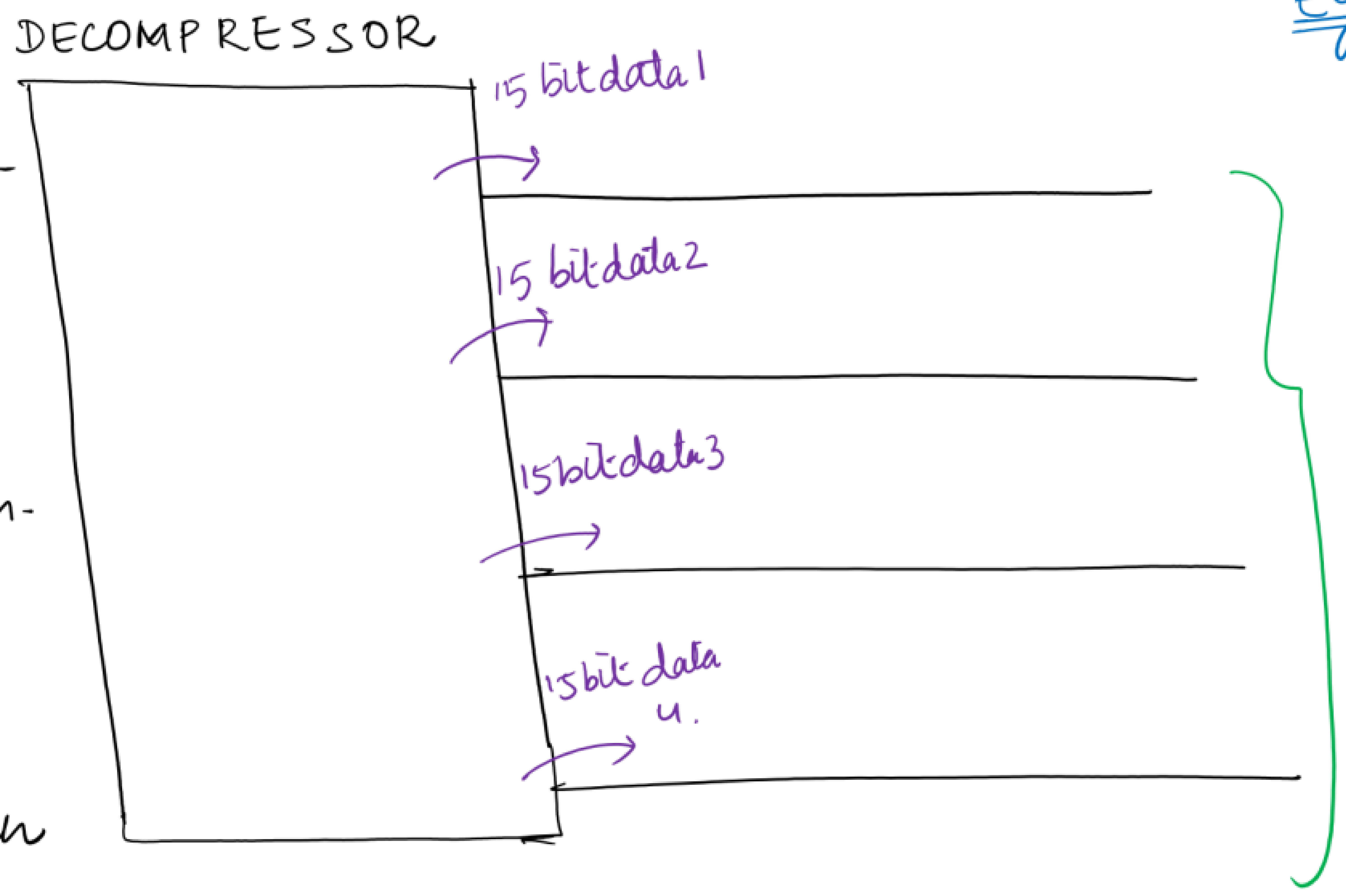
DECOMPRESSOR EXTRA

EXAMPLE :-

Eg:-
- internal chains = 4
- no. of ffs/chain = 15

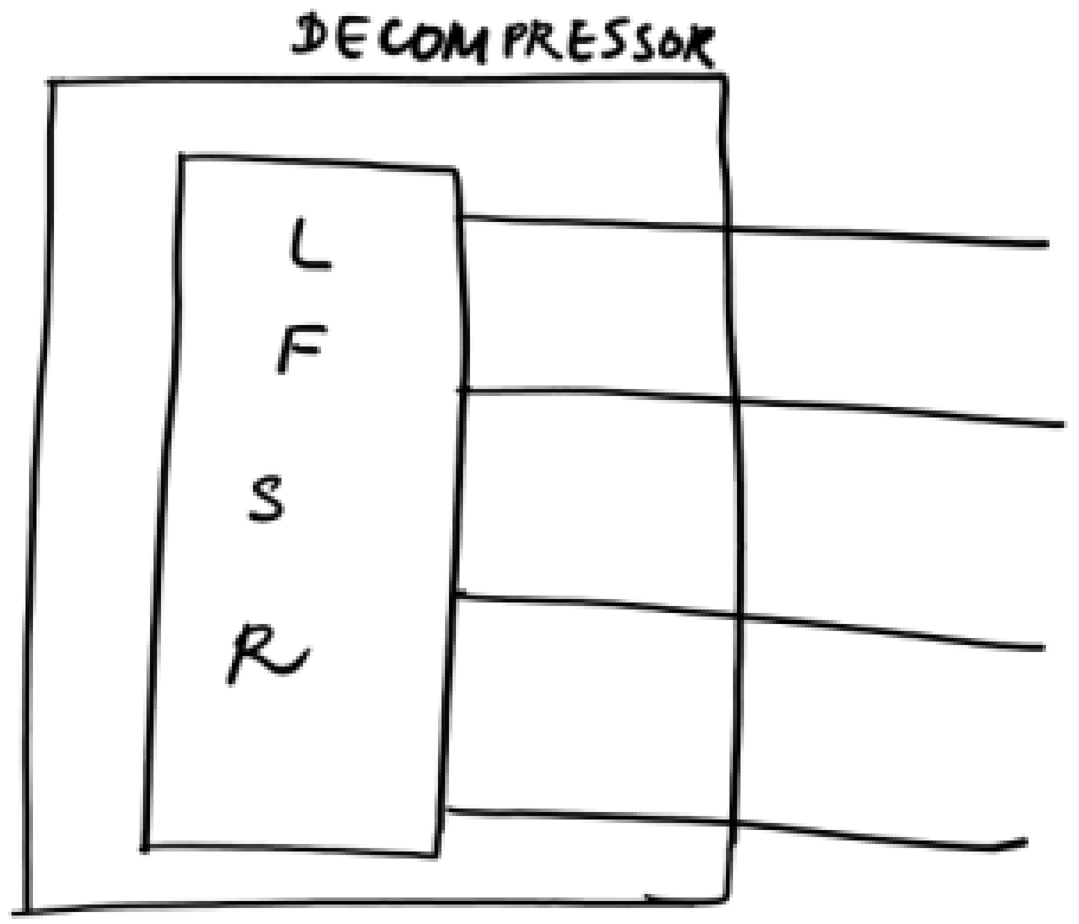
ROLE OF DECOMPRESSOR:-

To Apply a unique 15 bit pattern in each internal chain.



15 bit pattern
$\Rightarrow$ $\because$ there are 15 ffs in each chain

unique $\Rightarrow$ because the combo logic b/w 2 ffs will be different.
So pattern has to be loaded into the ffs inorder to test the respective combos.



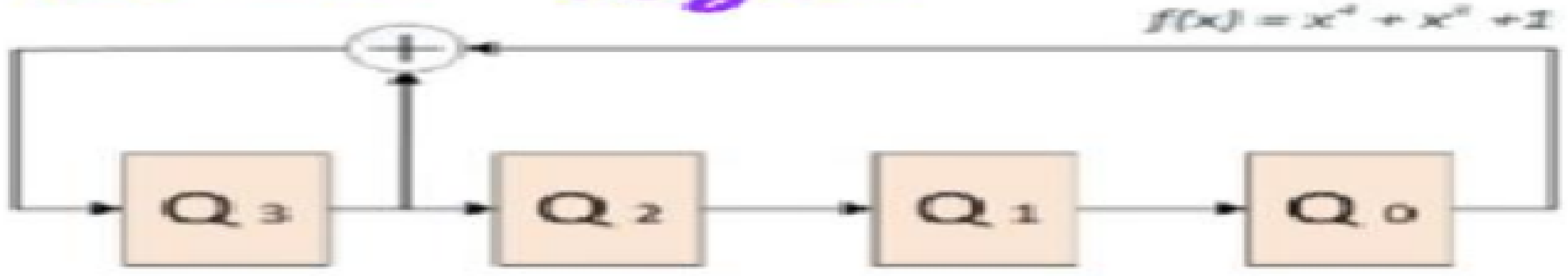
Given an initial value, the LFSR logic will start generating patterns (just like a counter counts)

But the patterns generated by the LFSR logic has a disadvantage. There will be diagonal relationship between adjacent bit streams (as shown in the diagram). Because of this, the tool will not be able to generate all possible combinations of patterns to detect all the faults.

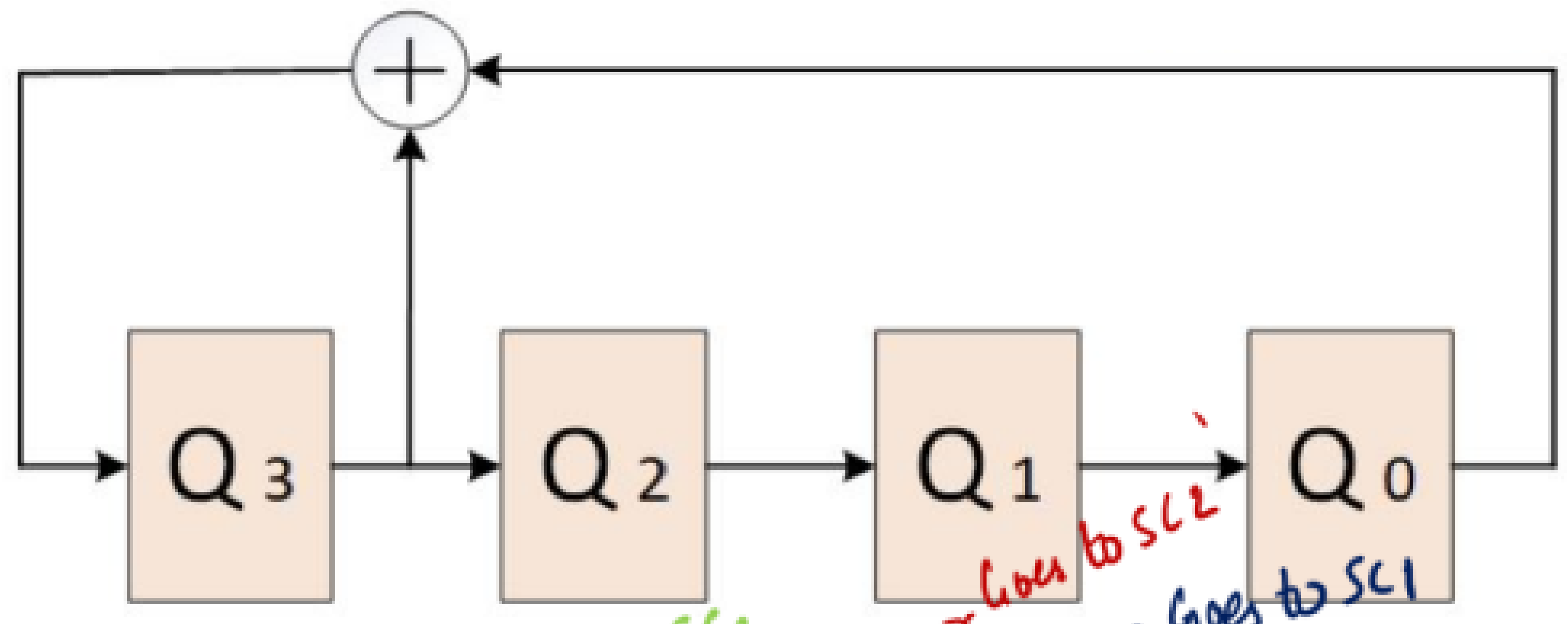
LFSR :-
Linear Feedback Shift Register

DISADVATAGE OF USING ONLY LFSR LOGIC :
Diagonal relationship b/w adj bit streams.
⇒ Tool will not be able to generate all possible combinations of pattens to detect the all the faults

LFSR Logic

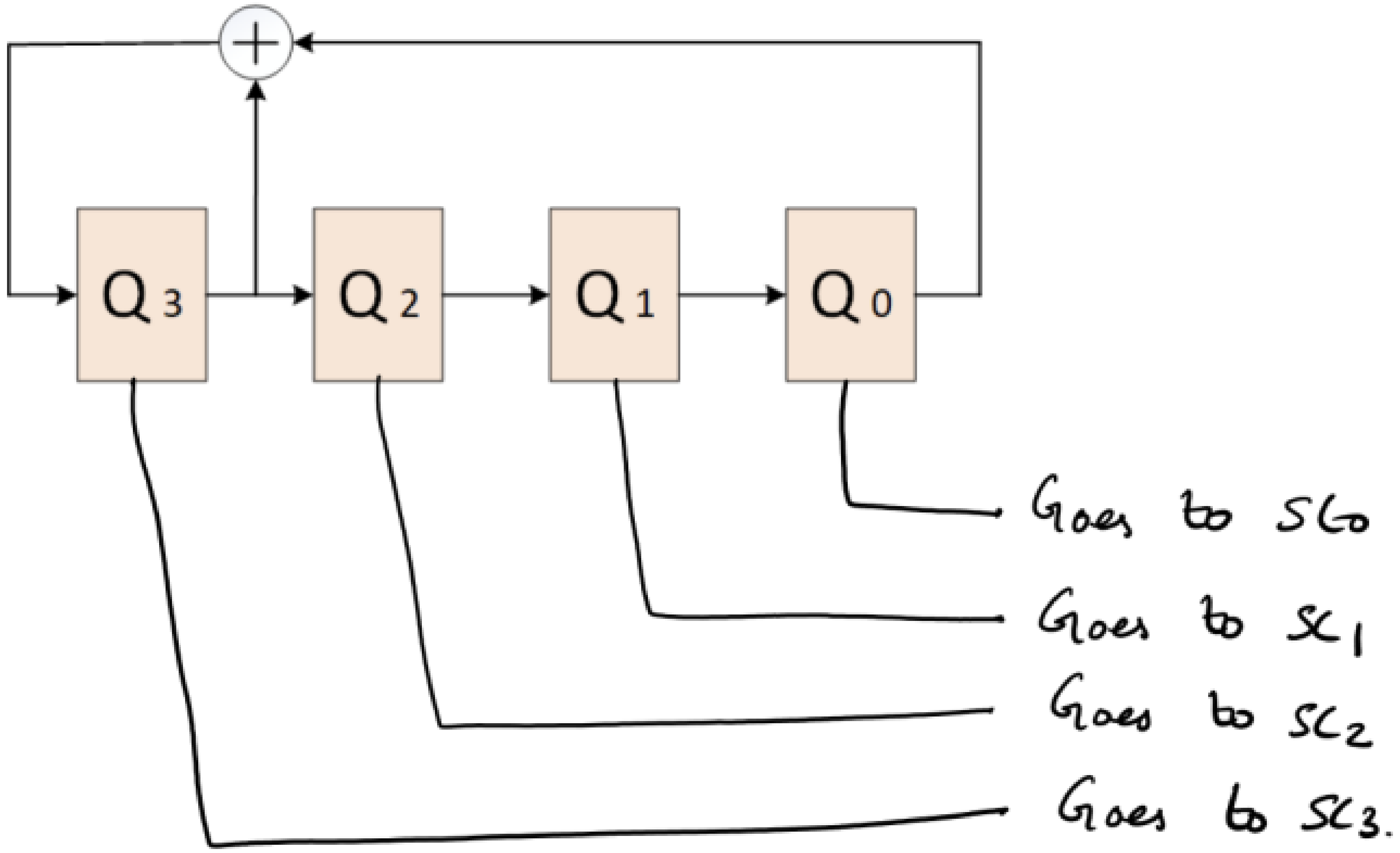


	Q_3	Q_2	Q_1	Q_0
15th clock cycle :	1	0	0	0
14th clock cycle :	0	0	0	1
13th clock cycle :	0	0	1	0
12th clock cycle :	0	1	0	0
11th clock cycle :	1	0	0	1
10th clock cycle :	0	0	1	1
9th clock cycle :	0	1	1	0
8th clock cycle :	1	1	0	1
7th clock cycle :	1	0	1	0
6th clock cycle :	0	1	0	1
5th clock cycle :	1	0	1	1
4th clock cycle :	0	1	1	1
3rd clock cycle :	1	1	1	1
2nd clock cycle :	1	1	1	0
1st clock cycle :	1	1	0	0
Initial value :	1	0	0	0

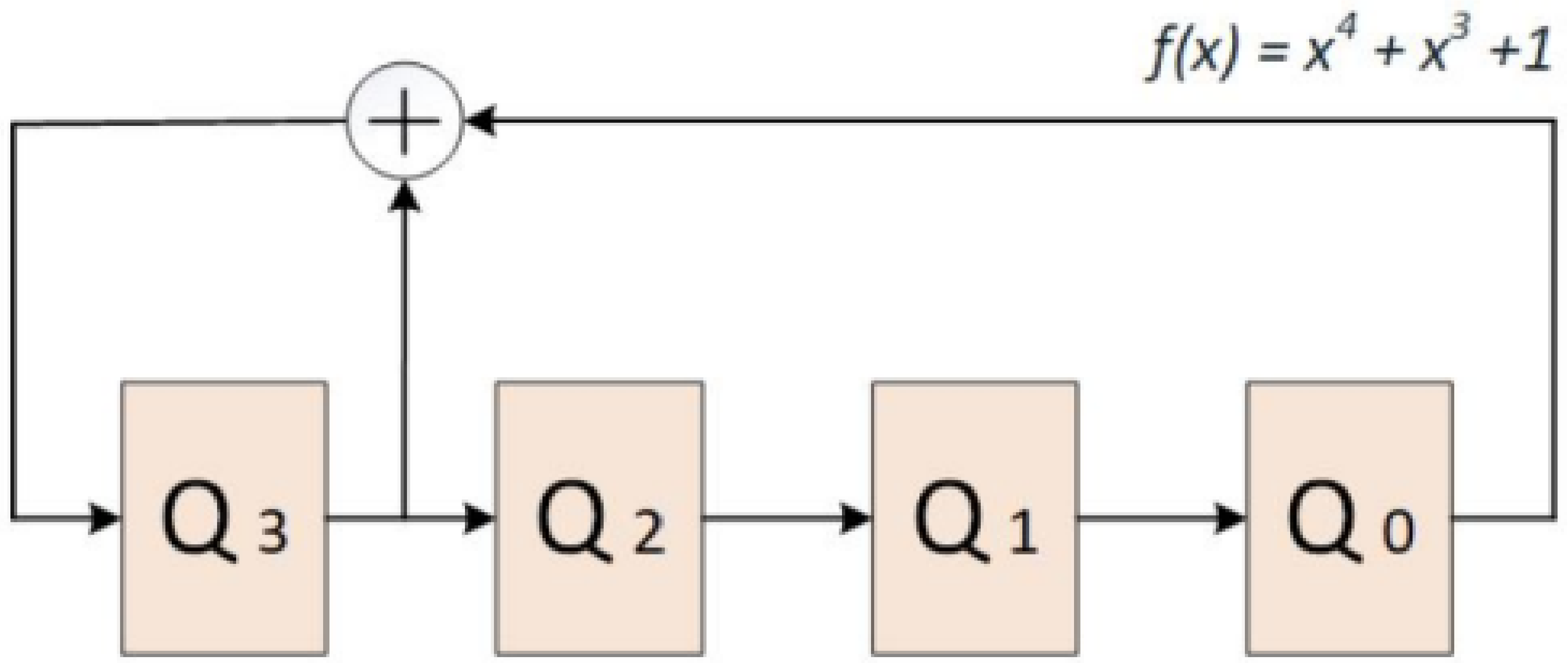


Goes to SC3 Goes to SC2 Goes to SC1 Goes to SC0

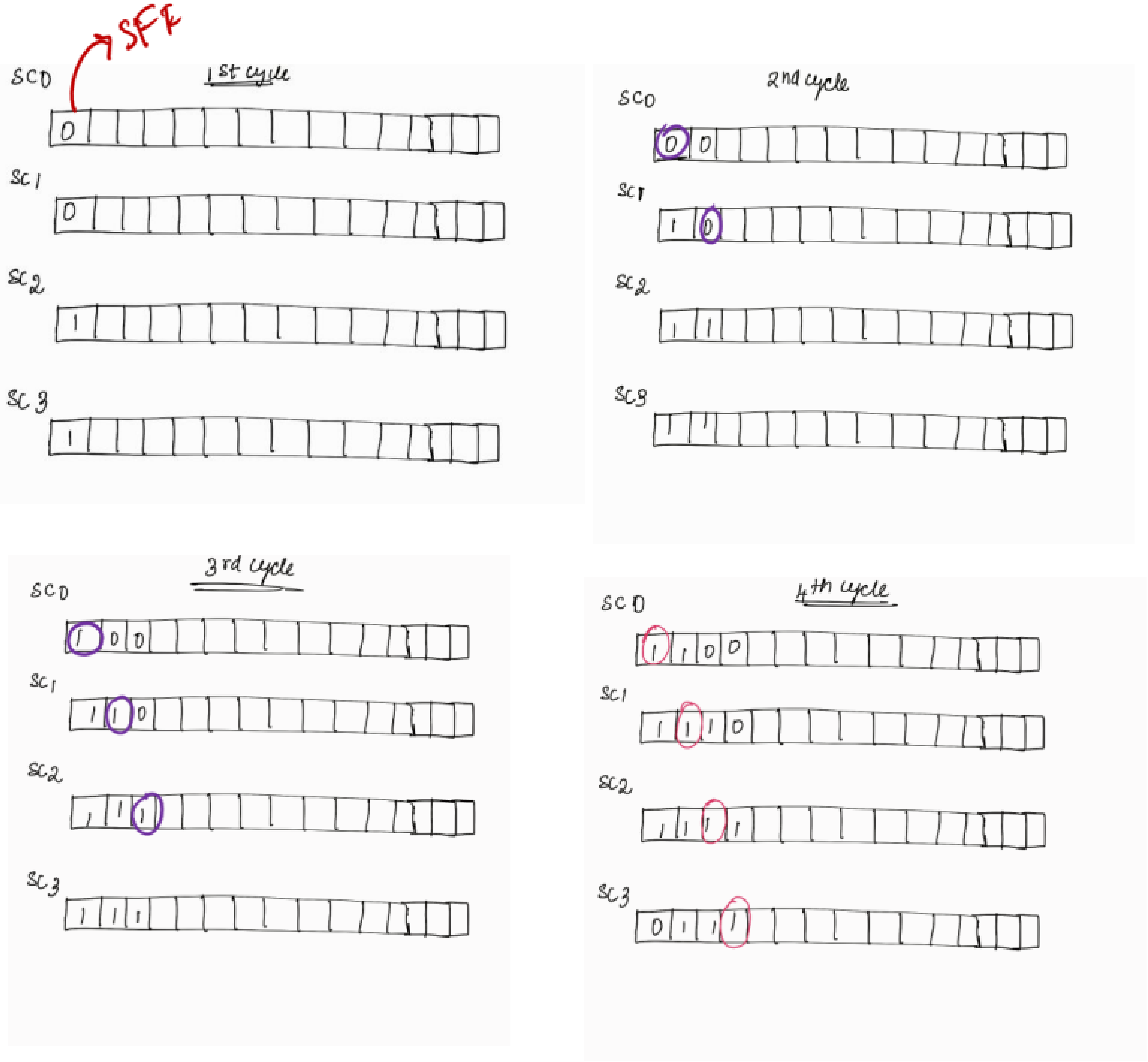
	Q_3	Q_2	Q_1	Q_0
15th clock cycle :	1	0	0	0
14th clock cycle :	0	0	0	1
13th clock cycle :	0	0	1	0
12th clock cycle :	0	1	0	0
11th clock cycle :	1	0	0	1
10th clock cycle :	0	0	1	1
9th clock cycle :	0	1	1	0
8th clock cycle :	1	1	0	1
7th clock cycle :	1	0	1	0
6th clock cycle :	0	1	0	1
5th clock cycle :	1	0	1	1
4th clock cycle :	0	1	1	1
3rd clock cycle :	1	1	1	1
2nd clock cycle :	1	1	1	0
1st clock cycle :	1	1	0	0
Initial value :	1	0	0	0



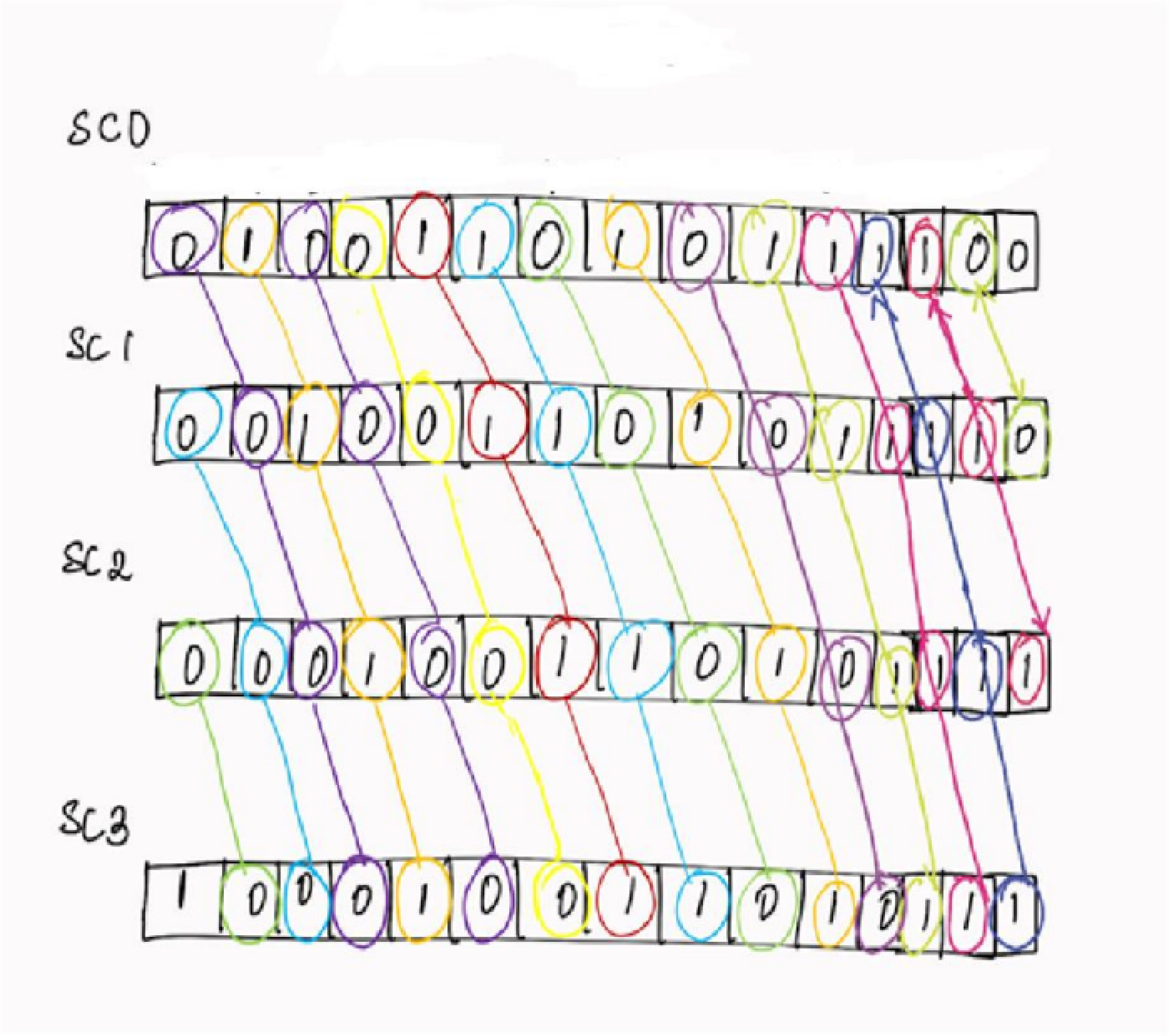
But the patterns generated by the LFSR logic has a disadvantage. There will be diagonal relationship between adjacent bit streams (as shown in the diagram). Because of this, the tool will not be able to generate all possible combinations of patterns to detect all the faults.

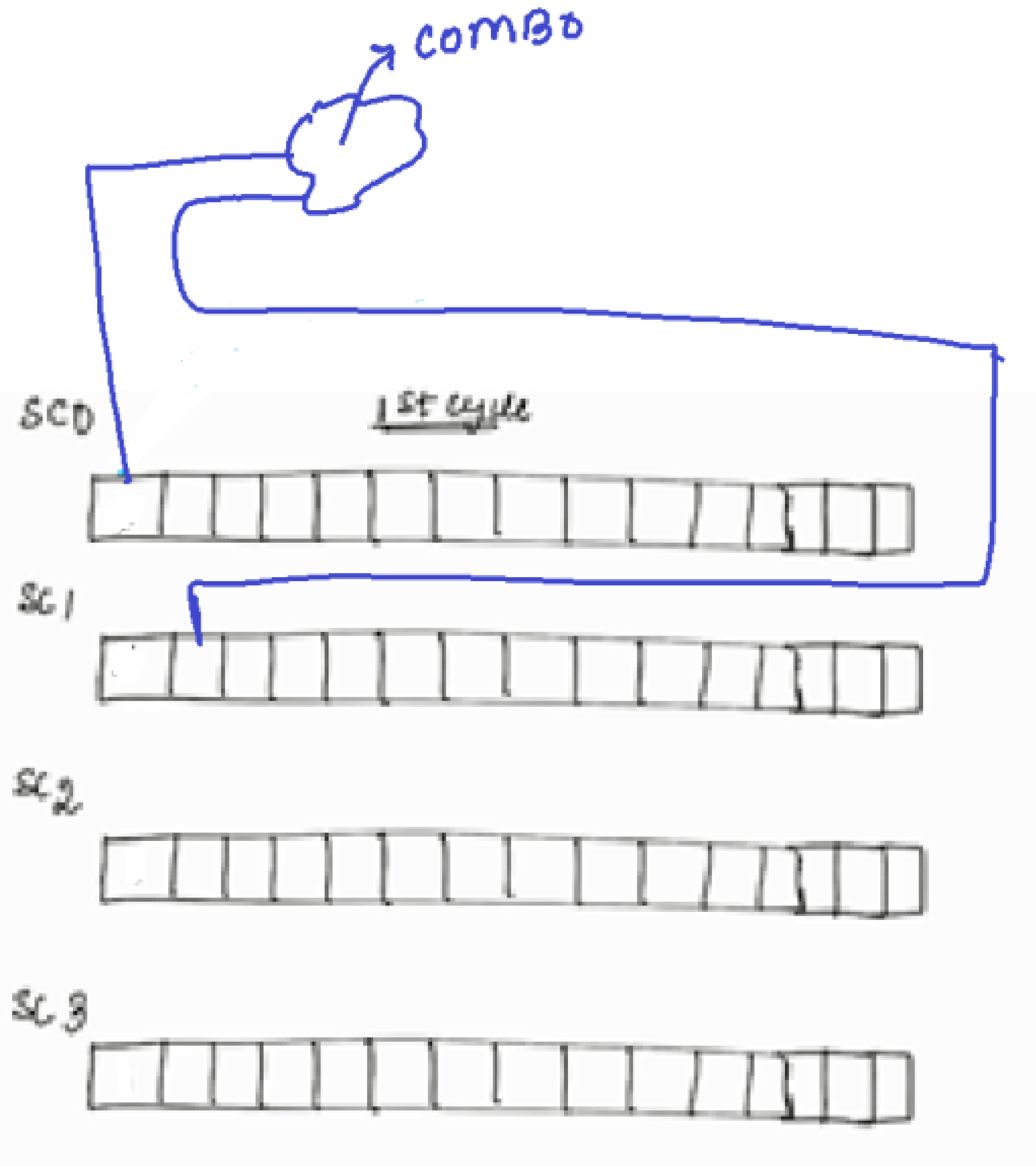


	Q_3	Q_2	Q_1	Q_0
15^{th} clock cycle :	1	0	0	0
14^{th} clock cycle :	0	0	0	1
13^{th} clock cycle :	0	0	1	0
12^{th} clock cycle :	0	1	0	0
11^{th} clock cycle :	1	0	0	1
10^{th} clock cycle :	0	0	1	1
9^{th} clock cycle :	0	1	1	0
8^{th} clock cycle :	1	1	0	1
7^{th} clock cycle :	1	0	1	0
6^{th} clock cycle :	0	1	0	1
5^{th} clock cycle :	1	0	1	1
4^{th} clock cycle :	0	1	1	1
3^{rd} clock cycle :	1	1	1	1
2^{nd} clock cycle :	1	1	1	0
1^{st} clock cycle :	1	1	0	0
Initial value :	1	0	0	0



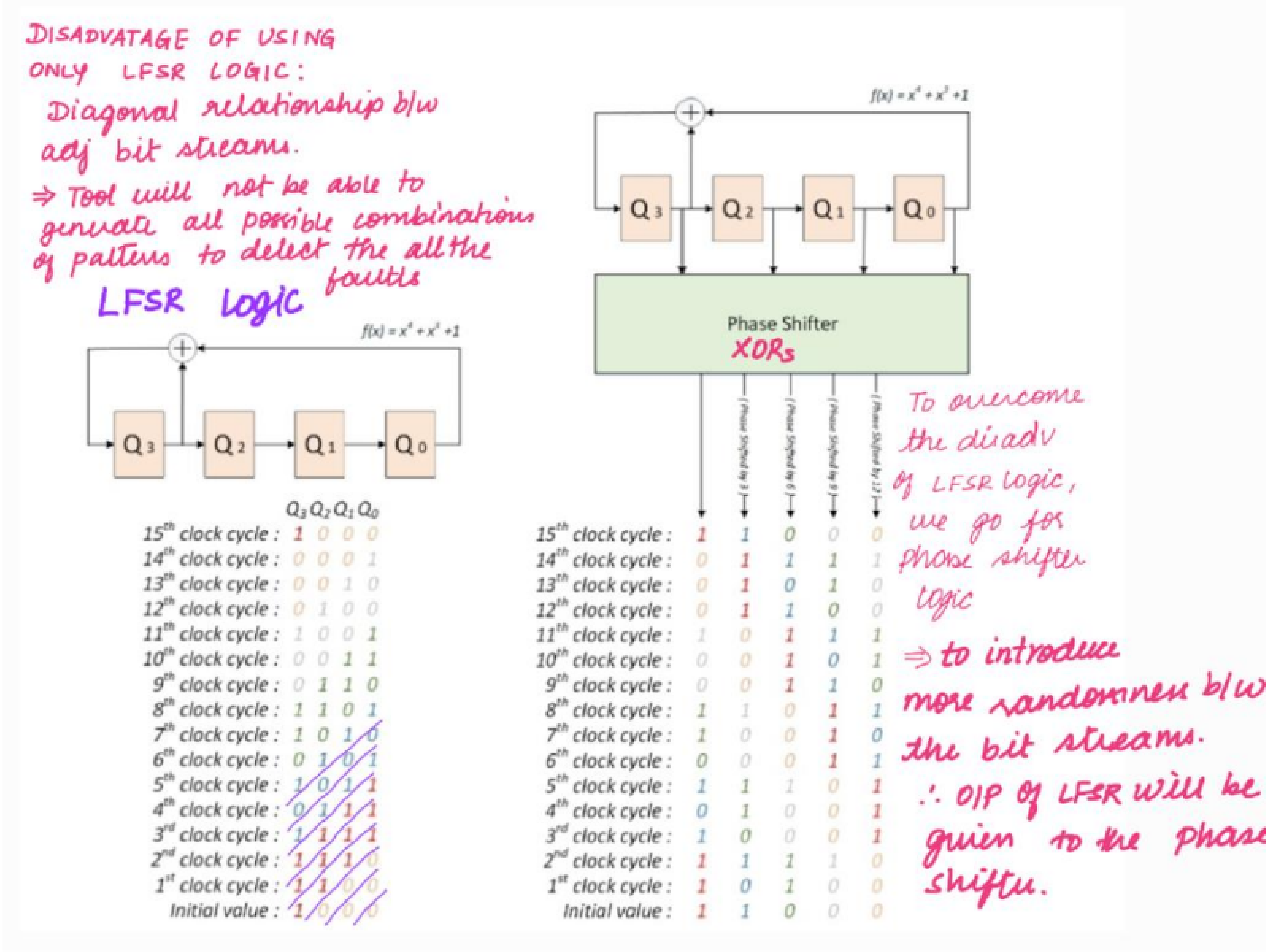
After 15 cycles

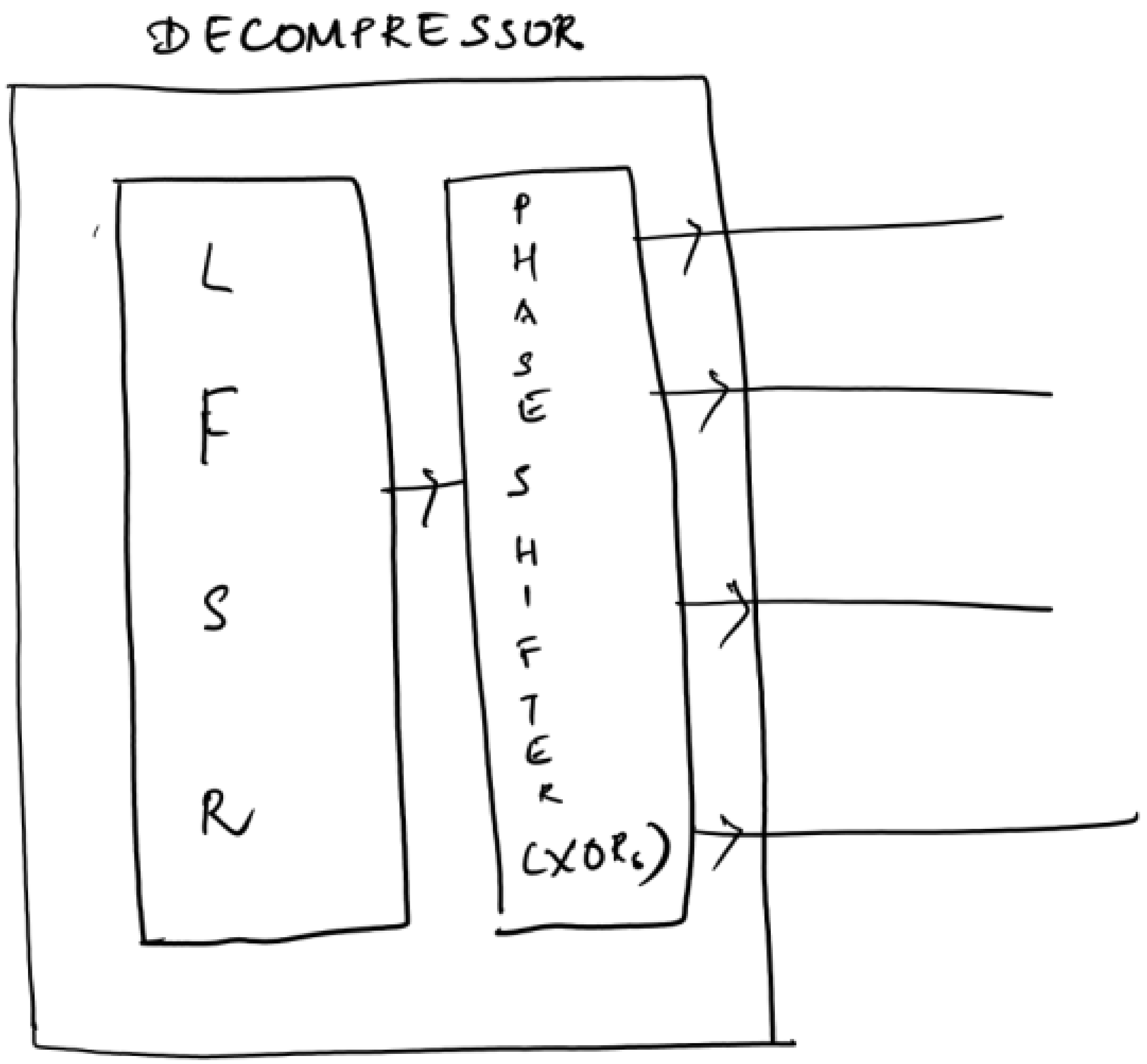




Suppose, Scan chain 0's first flop and Scan chain 1's second flop goes as an input to some combo as shown in the figure, the tool will not be able to generate all possible combinations of patterns to detect all the faults in that combo.
It is because, as seen from the above demonstration, we are unable to bring 2 different values in those 2 flops in any cycle.
We are unable to bring 2 different values in those 2 flops in any cycle because, there will be a diagonal relationship between adjacent bit streams in the patterns generated by the LFSR logic.

So in order to introduce more randomness between the bit streams, output of LFSR logic will be given to the phase shifter logic.





But even after adding the phase shifter logic, the tool might not be able to generate all possible combinations of patterns to the detect the faults.

So to further increase the randomness to the patterns, patterns will also be given from external channels.

Eg:- Given a 15 bit data as an i/p to the decomp, the decomp will be able to generate a unique 15 bit data for each of the scan chain and the values coming out of the decompressor will be random enough to generate all possible combination of patterns to detect most of the faults.